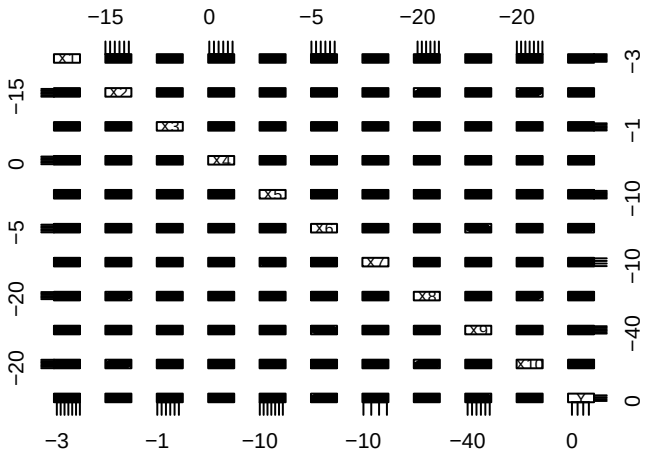
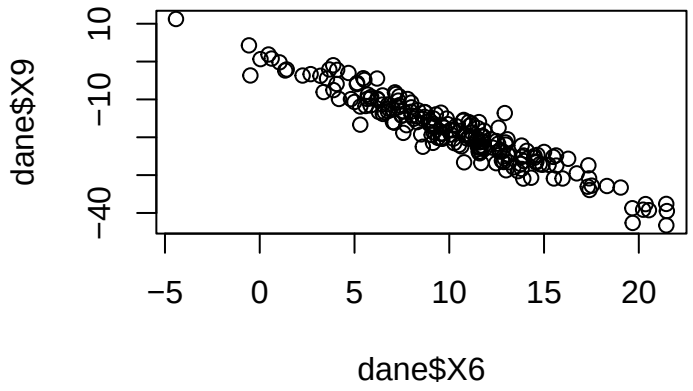
Sprawozdanie 2

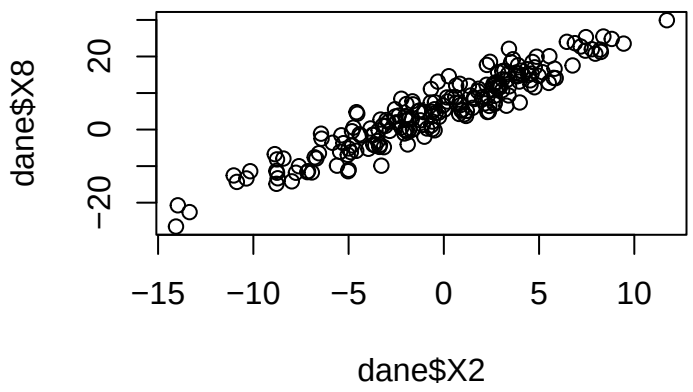
Michał Frąckowiak
275951

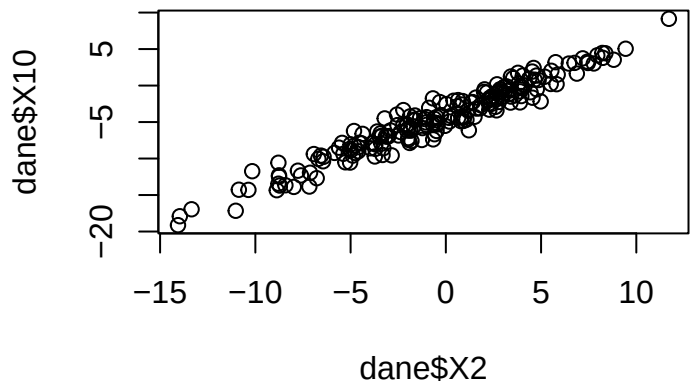
26 marca 2026

Spis treści


X1
X2
X3
X4
X5
X6
X7
X8
X9
−15
0
−5
−20
−20
−3
−1
−10
−10
−40
0


dane$X9
10
−10
−40
−5
0
5
10
15
20
dane$X6


dane$X8
20
0
−20
−15
−10
−5
0
5
10
dane$X2

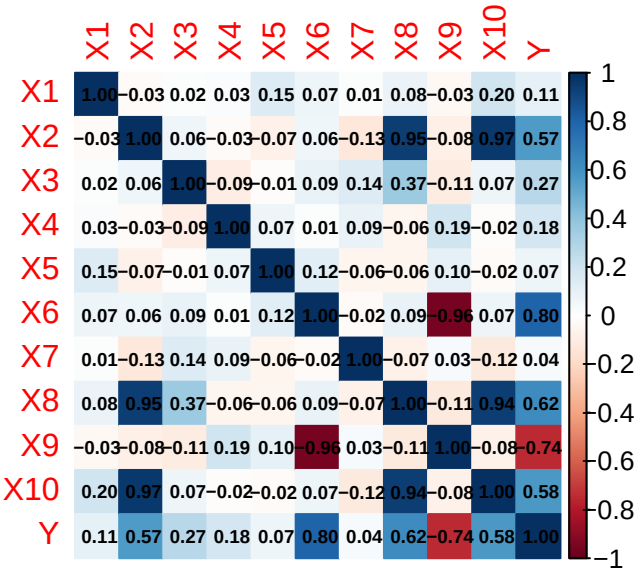



```
##          X1     X2     X3     X4     X5     X6     X7     X8     X9    X10
## X1   1.000 -0.026  0.018  0.025  0.147  0.072  0.010  0.082 -0.032  0.198
## X2  -0.026  1.000  0.064 -0.030 -0.071  0.061 -0.130  0.946 -0.082  0.969
## X3   0.018  0.064  1.000 -0.094 -0.013  0.091  0.145  0.367 -0.108  0.071
## X4   0.025 -0.030 -0.094  1.000  0.072  0.013  0.090 -0.055  0.194 -0.016
## X5   0.147 -0.071 -0.013  0.072  1.000  0.124 -0.058 -0.056  0.105 -0.024
## X6   0.072  0.061  0.091  0.013  0.124  1.000 -0.024  0.091 -0.955  0.071
## X7   0.010 -0.130  0.145  0.090 -0.058 -0.024  1.000 -0.073  0.031 -0.119
## X8   0.082  0.946  0.367 -0.055 -0.056  0.091 -0.073  1.000 -0.112  0.942
## X9  -0.032 -0.082 -0.108  0.194  0.105 -0.955  0.031 -0.112  1.000 -0.078
## X10  0.198  0.969  0.071 -0.016 -0.024  0.071 -0.119  0.942 -0.078  1.000
## Y    0.107  0.568  0.272  0.184  0.073  0.805  0.036  0.622 -0.741  0.577
##           Y
## X1    0.107
## X2    0.568
## X3    0.272
## X4    0.184
## X5    0.073
## X6    0.805
## X7    0.036
## X8    0.622
## X9   -0.741
## X10   0.577
## Y     1.000
```

```
##
## Call:
## lm(formula = Y ~ ., data = dane)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.19580 -0.63487  0.05182  0.61069  2.90932
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.36633    0.61843  -0.592 0.554321
## X1           1.31223    0.33611   3.904 0.000131 ***
## X2           2.48179    0.59755   4.153 4.96e-05 ***
## X3           3.64392    0.84810   4.297 2.77e-05 ***
## X4           4.41298    0.19230  22.949  < 2e-16 ***
## X5           0.48266    0.19489   2.477 0.014142 *
## X6           5.10263    0.38470  13.264  < 2e-16 ***
## X7           1.05125    0.02515  41.804  < 2e-16 ***
## X8           0.83739    0.27864   3.005 0.003013 **
## X9          -0.43897    0.19274  -2.278 0.023875 *
## X10         -0.15381    0.13267  -1.159 0.247771
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9259 on 189 degrees of freedom
## Multiple R-squared:  0.9994, Adjusted R-squared:  0.9994
## F-statistic: 3.077e+04 on 10 and 189 DF,  p-value: < 2.2e-16
```

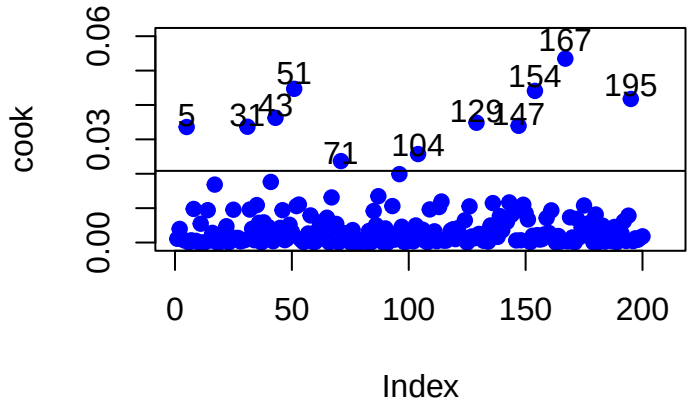
```
## (Intercept)          X1          X2          X3          X4          X5
##  -0.3663314   1.3122296   2.4817921   3.6439204   4.4129815   0.4826617
##          X6          X7          X8          X9         X10
##   5.1026333   1.0512518   0.8373855  -0.4389655  -0.1538112
```

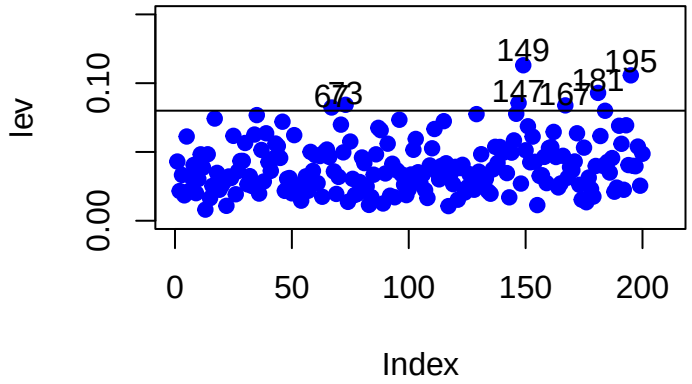


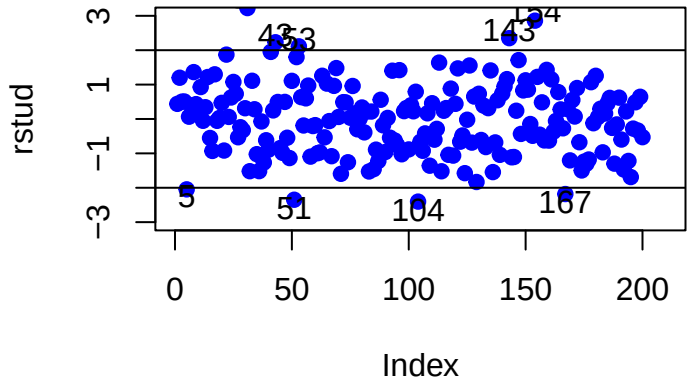

```
## [1] 0.9993536
## [1] 0.999386
```

```
##          X1          X2          X3          X4          X5          X6          X7
##    5.673484   91.309968    1.056029   28.080197   34.600497  713.894340    1.066055
##          X9         X10
##  736.536504   94.608677
##        X1        X2        X3        X4        X5        X6        X7       X10
```

```
##  5.661631 91.259379  1.052635  1.031352  1.070885  1.037012  1.062009 94.552938
##        X1        X2        X3        X4        X5        X6        X7
## 1.026188 1.035339 1.050014 1.026707 1.052000 1.032656 1.059852
```






rstud
3
1
−1
−3
143
5
51
104
167
0
50
100
150
200
Index